WORKED SOLUTION

STEP II 1994, Question 13

Mechanics • Exponential distribution

STEP II 1994, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 9.11, *Collecting the Set*: https://beyondstep.keithmcnulty.org/ch09/coupon-collector.html

The key idea

This is the classic "coupon collector" problem, and despite the database tag it is pure probability. Split the long wait for a full set into n shorter waits: the wait for the first new card, then the second, and so on. Each short wait is geometric, so its mean is easy, and linearity of expectation adds them up. The total is n times a harmonic sum, and the integral inequality is there to tell us how big that sum is.

Solution

Waiting for a new card

We model each packet as containing one of the n cards, each equally likely with probability $\frac{1}{n}$, independently of every other packet.

Suppose I already hold r different cards, where $0 \leq r \leq n-1$. A packet gives a new card with probability

$$p = \frac{n-r}{n} > 0,$$

and an old card with probability $q = 1 - p = \frac{r}{n}$. Let X be the number of packets I open up to and including the first new card. Because the packets are independent, $X = k$ means $k-1$ old cards followed by a new one, so

$$\mathrm{P}(X = k) = q^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

Hence

$$\mathrm{E}(X) = \sum_{k=1}^{\infty} kpq^{k-1} = p\,(1-q)^{-2} = \frac{p}{p^2} = \frac{1}{p},$$

using $\sum_{k\geq 1} kq^{k-1} = (1-q)^{-2}$ for $|q| < 1$ (differentiate the geometric series, or expand $(1-q)^{-2}$ binomially). Here $0 \leq q < 1$ because $r \leq n-1$. So

$$\mathrm{E}(X) = \frac{n}{n-r},$$

as required.

(A slicker route: the first packet either gives a new card, and we stop, or with probability q it puts us back where we started, so $\mathrm{E}(X) = 1 + q\,\mathrm{E}(X)$, giving $\mathrm{E}(X) = 1/(1-q) = 1/p$. This assumes $\mathrm{E}(X)$ is finite, which the series above confirms.)

The integral inequality

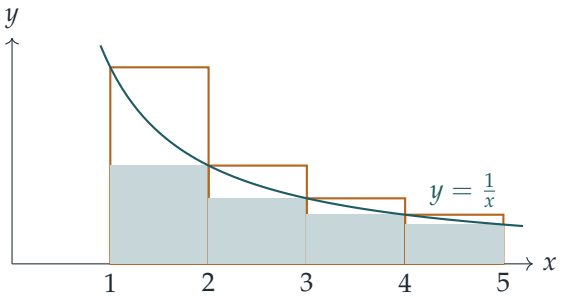
Let $r \geq 1$. For $r \leq x \leq r+1$ we have

$$\frac{1}{r+1} \leq \frac{1}{x} \leq \frac{1}{r},$$

because $1/x$ is decreasing for $x > 0$. Integrating over an interval of length 1 keeps these inequalities, so

$$\frac{1}{r+1} \leq \int_r^{r+1} \frac{1}{x}\,\mathrm{d}x \leq \frac{1}{r}.$$

In the picture, the area under $y = 1/x$ between $x = r$ and $x = r+1$ lies between the short rectangle of height $\frac{1}{r+1}$ and the tall rectangle of height $\frac{1}{r}$.



(The vertical scale is stretched for clarity.) The shaded rectangles sit under the curve; the outlined ones sit above it.

Now add the inequality for $r = 1, 2, \ldots, n-1$, where $n \geq 2$. The integrals join up into one:

$$\sum_{r=1}^{n-1} \frac{1}{r+1} \leq \int_1^n \frac{1}{x}\,\mathrm{d}x \leq \sum_{r=1}^{n-1} \frac{1}{r}.$$

The middle term is $\ln n - \ln 1 = \ln n$, and in the left-hand sum we relabel $r+1$ as r, which runs from 2 to n. So

$$\sum_{r=2}^{n} \frac{1}{r} \leq \ln n \leq \sum_{r=1}^{n-1} \frac{1}{r},$$

as required.

Collecting the whole set

Let X_r be the number of packets opened while I hold exactly r different cards, ending with the packet that gives card number $r+1$. The total number of packets needed for the full set is

$$T = X_0 + X_1 + \cdots + X_{n-1}.$$

Ask yourself The X_r depend on what happened earlier. Can we still just add their expectations?

Yes. Linearity of expectation, $\mathrm{E}(A+B) = \mathrm{E}(A) + \mathrm{E}(B)$, needs no independence at all. (In fact, given the history so far, X_r is geometric with $p = \frac{n-r}{n}$ regardless, so the first part applies to each X_r.) Therefore

$$\mathrm{E}(T) = \sum_{r=0}^{n-1} \frac{n}{n-r} = n\left(\frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{1}\right) = n\sum_{k=1}^{n}\frac{1}{k},$$

writing $k = n - r$.

Write $H_n = \sum_{k=1}^{n} \frac{1}{k}$. The two halves of the inequality give

$$H_n = 1 + \sum_{k=2}^{n}\frac{1}{k} \le 1 + \ln n, \qquad H_n = \sum_{k=1}^{n-1}\frac{1}{k} + \frac{1}{n} \ge \ln n + \frac{1}{n}.$$

So

$$n\ln n + 1 \le \mathrm{E}(T) \le n\ln n + n.$$

The gap between the bounds is less than n, which is tiny next to $n\ln n$ when n is large (their ratio $1/\ln n \to 0$). So for large n I should expect to eat roughly

$$n\ln n \text{ packets.}$$

For example, 100 mathematicians would mean about 460 packets of Cruncho.

Checks

- $r = 0$: the first packet is always new, and $n/(n-0) = 1$. $r = n-1$: one specific card is missing, which turns up with probability $\frac{1}{n}$, so we expect n packets, matching $n/1$.
- $n = 2$: $\mathrm{E}(T) = 2(1 + \frac{1}{2}) = 3$, and the bounds give $2\ln 2 + 1 \approx 2.39$ and $2\ln 2 + 2 \approx 3.39$.
- A simulation of 10^5 complete collections with $n = 20$ gave an average of 71.9 packets, against $20H_{20} \approx 71.95$; the bounds are 60.9 and 79.9.

Common slips

- Summing the integral inequality from $r = 0$: the bound $\frac{1}{r}$ fails there, and $\int_0^1 \frac{1}{x}\,\mathrm{d}x$ diverges. The sum must start at $r = 1$.

- Thinking the X_r must be independent before adding expectations. Linearity needs no independence.
- Answering $\ln n$ or n: the expected number of packets grows like $n \ln n$, faster than n.

Beyond STEP

The difference $H_n - \ln n$ decreases to a limit, the Euler–Mascheroni constant $\gamma \approx 0.5772$, so a sharper answer is $\mathrm{E}(T) \approx n(\ln n + \gamma)$; for $n = 100$ that gives about 519 packets rather than 460. Comparing sums with integrals like this is the integral test for series, met early in a first analysis course. The coupon collector problem itself reappears in the analysis of algorithms and in the study of how many random shuffles mix a deck. Grimmett and Welsh, *Probability: An Introduction*, is a good place to see geometric waiting times and linearity of expectation used together.

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